Assignment 04

Data Preprocessing .

EUID = gg0640

Cleansing 1 if n=0+1 =1

Here I had practiced all the 6 steps in the data preprocessing.

But as mentioned in the Cleansing 1

Goal: Our goal is to handle the missing values and in this case, we had represented the missing values with a Nan.

This means that that it is “not a number “ .

1. By reindexing our rows, the rows are increased for a specific amount of values which raises missing values and those are replaced and represented using the NaN values .
2. Later in the cleansing steps , we dive into checking if there are any null values with a Boolean output , filling the null values with zeros and replacing the null values with a different number scalar values and so on .
3. There are also the fill Na values in a forward or backward type which should be mentioned in the “method= pad” etc.

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```python
# canvas login id = gg0640
# cleansing 1 if 0+1=1 ;
# practised every technique
import pandas as pd
import numpy as np

df_variable = pd.DataFrame(np.random.randn(5,4),index=[1,3,5,7,9],columns=['a','b','c','d'])
df_variable = df_variable.reindex([1,2,3,4,5,6,7,8])

print(df_variable)
```

```
          a         b         c         d
1 -0.427676 -1.429306 -0.774875 -0.944359
2       NaN       NaN       NaN       NaN
3  0.629715  1.017324 -1.068677  0.919546
4       NaN       NaN       NaN       NaN
5 -1.067656  0.696967 -0.057127  1.113677
6       NaN       NaN       NaN       NaN
7  1.212217 -0.168888  0.023145 -0.947534
8       NaN       NaN       NaN       NaN
```

```python
# cleansing 2
print(df_variable['a'].isnull())
```

```
1    False
2     True
3    False
4     True
5    False
6     True
7    False
8     True
Name: a, dtype: bool
```

Screenshot

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```python
# cleansing 3
df_var = df_variable
print(df_var.fillna(0))
```

```
          a         b         c         d
1 -0.427676 -1.429306 -0.774875 -0.944359
2  0.000000  0.000000  0.000000  0.000000
3  0.629715  1.017324 -1.068677  0.919546
4  0.000000  0.000000  0.000000  0.000000
5 -1.067656  0.696967 -0.057127  1.113677
6  0.000000  0.000000  0.000000  0.000000
7  1.212217 -0.168888  0.023145 -0.947534
8  0.000000  0.000000  0.000000  0.000000
```

```python
# cleansing 4
# filling null values here with fill method forward condtion
df_var1=df_variable
print(df_var1.fillna(method='pad'))
```

```
          a         b         c         d
1 -0.427676 -1.429306 -0.774875 -0.944359
2 -0.427676 -1.429306 -0.774875 -0.944359
3  0.629715  1.017324 -1.068677  0.919546
4  0.629715  1.017324 -1.068677  0.919546
5 -1.067656  0.696967 -0.057127  1.113677
6 -1.067656  0.696967 -0.057127  1.113677
7  1.212217 -0.168888  0.023145 -0.947534
8  1.212217 -0.168888  0.023145 -0.947534
```

```python
# cleansing 5
# dropping null vales
df_var2=df_variable
print(df_var2.dropna())
```

Screenshot

```
          a         b         c         d
```